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| United States Patent Application Publication | 20250285988 |
| Kind Code | A1 |
| Publication Date | September 11, 2025 |
| Inventor(s) | Wang; Bau-Ming et al. |


## DOPED WELL AND ALIGNMENT MARK FOR SEMICONDUCTOR DEVICES


### Abstract

In an embodiment, a semiconductor device includes a semiconductor substrate including a first fin and a second fin; a first well, a second well and an alignment mark in the semiconductor substrate; and an isolation layer over the first fin, the second fin, the semiconductor substrate, and the alignment mark. The first fin is in the first well, and the first well is doped with a first dopant having a first conductivity type. The second fin is in the second well, and the second well is doped with a second dopant having a second conductivity type different from the first conductivity type. The alignment mark includes a lower portion formed of a first material and an upper portion formed of a second material different from the first material.



**Inventors:** **Wang; Bau-Ming (Kaohsiung City, TW), Chen; Liang-Yin (Hsinchu, TW), Chen; Chien-Hao (Chuangwei Township, TW)**

**Applicant:** **Taiwan Semiconductor Manufacturing Co., Ltd.** (Hsinchu, TW)



**Family ID:** **1000007744610**

**Appl. No.:** **18/599695**

**Filed:** **March 08, 2024**

### Publication Classification

**Int. Cl.:** **H01L23/544** (20060101); **H01L21/266** (20060101); **H01L29/06** (20060101); **H01L29/423** (20060101); **H01L29/775** (20060101); **H01L29/786** (20060101)

**U.S. Cl.:**

CPC **H01L23/544** (20130101); **H01L21/266** (20130101); **H10D30/43** (20250101); **H10D30/6735** (20250101); **H10D30/6757** (20250101); **H10D62/121** (20250101); H01L2223/54426 (20130101)

## Background/Summary

BACKGROUND

[0001] Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

[0002] The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, as the minimum features sizes are reduced, additional problems arise that should be addressed.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. **1** illustrates an example of a nanostructure field-effect transistor (nano-FET) in a three-dimensional view, in accordance with some embodiments.

[0005] FIGS. **2**, **3**A, **3**B and **4** to **9** illustrate intermediate stages in manufacturing wells and alignment marks of nano-FETs, in accordance with some embodiments.

[0006] FIGS. **10**, **11**A, **11**B, **12**A, **12**B, **13**, **14**A, **14**B, **15**A, **15**B, **16**A, **16**B, **17**A, **17**B, **18**A, **18**B, **19**A, **19**B, **19**C, **20**A, **20**B, **20**C, **20**D, **21**A, **21**B, **21**C, **22**A, **22**B, **23**A, **23**B, **24**A, **24**B, **25**A, **25**B, **26**A, **26**B, **26**C, **26**A, **26**B, **26**C, **27**A, **27**B, **27**C, **28**A, **28**B, **28**C, **28**D, **29**A, **29**B, and **29**C illustrate various intermediate stages in the manufacturing of nano-FETs, in accordance with some embodiments.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0009] As discussed in greater detail below, embodiments of the present disclosure describe a dopant implantation process to form p-wells and/or n-wells in a substrate, which may be used to form a transistor (e.g., nano-FETs, fin field effect transistors (FinFETs), planar transistors, or the like). The techniques described herein include using hard implantation masks in implanting the p-wells and/or n-wells. The hard implantation masks may prevent or reduce shrinkage in a high temperature process and can improve the dopant profile of the p-wells and/or n-wells in the lateral direction, thereby improving the junction of neighboring p-wells and n-wells, which may be desirable for transistors with small critical dimensions of p-wells and n-wells. In some embodiments, the materials of the hard masks may also form alignment marks with or without dopants therein, which may provide the alignment marks various adjustable characteristics that can be sensitive to particular processing equipment. Embodiments are described below in a particular context, a die comprising nano-FETs. Various embodiments may be applied, however, to dies comprising other types of transistors, such as FinFETs, planar transistors, or the like, in lieu of or in combination with the nano-FETs.

[0010] FIG. **1** illustrates an example of nano-FETs (e.g., nanosheet FETs, or the like) in a three-dimensional view, in accordance with some embodiments. The nano-FETs comprise nanostructures **55** (e.g., nanosheets or the like) over fins **66** on a substrate **10** (e.g., a semiconductor substrate), wherein the nanostructures **55** act as channel regions for the nano-FETs. The nanostructures **55** may include p-type nanostructures, n-type nanostructures, or a combination thereof. Isolation regions **68** are disposed between adjacent fins **66**, which may protrude above and from between neighboring isolation regions **68**. Although the isolation regions **68** are described/illustrated as being separate from the substrate **10**, as used herein, the term “substrate” may refer to the semiconductor substrate alone or a combination of the semiconductor substrate and the isolation regions. Additionally, although a bottom portion of the fins **66** are illustrated as being single, continuous materials with the substrate **10**, the bottom portion of the fins **66** and/or the substrate **10** may comprise a single material or a plurality of materials. In this context, the fins **66** refer to the portion extending between the neighboring isolation regions **68**.

[0011] Gate dielectric layers **100** are over top surfaces of the fins **66** and along top surfaces, sidewalls, and bottom surfaces of the nanostructures **55**. Gate electrodes **102** are over the gate dielectric layers **100**. Epitaxial source/drain regions **92** are disposed on the fins **66** on opposing sides of the gate dielectric layers **100** and the gate electrodes **102**.

[0012] FIG. **1** further illustrates reference cross-sections that are used in later figures. Cross-section A-A′ is along a longitudinal axis of a gate electrode **102** and in a direction, for example, perpendicular to the direction of current flow between the epitaxial source/drain regions **92** of a nano-FET. Cross-section B-B′ is substantially perpendicular to cross-section A-A′ and is substantially parallel to a longitudinal axis of a fin **66** of the nano-FET and in a direction of, for example, a current flow between the epitaxial source/drain regions **92** of the nano-FET, within process variations. Cross-section C-C′ is parallel to cross-section A-A′ and extends through epitaxial source/drain regions of the nano-FETs. Subsequent figures refer to these reference cross-sections for clarity.

[0013] Some embodiments discussed herein are discussed in the context of nano-FETs formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in other devices, such as planar FETs or FinFETs.

[0014] FIG. **2** through **29**C illustrate various intermediate stages in the manufacturing of nano-FETs, in accordance with some embodiments. FIGS. **2**, **3**A, **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11**A, **12**A, **13**, **14**A, **21**A, **22**A, **23**A, **24**A, **25**A, **26**A, **27**A, **28**A, and **29**A illustrate reference cross-section A-A′ illustrated in FIG. **1**. FIGS. **14**B, **15**B, **16**B, **17**B, **18**B, **19**B, **19**C, **20**B, **20**D, **21**B, **22**B, **23**B, **24**B, **25**B, **26**B, **27**B, **28**B, and **29**B illustrate reference cross-section B-B′ illustrated in FIG. **1**. FIGS.

**15**A, **16**A, **17**A, **18**A, **19**A, **20**A, **20**C, **21**C, **26**C, **27**C, **28**C, and **29**C illustrate reference cross-section C-C′ illustrated in FIG. **1**. FIGS. **11**B, **12**B, and **28**D illustrate reference a cross-section that is parallel to reference cross-section A-A′ or B-B′ albeit in a different portion of the structure, such as a portion where an alignment mark is formed.

[0015] FIGS. **2** to **9** illustrate methods of forming wells and alignment marks in a substrate, in accordance with some embodiments. Additional processing is contemplated by the present disclosure. Referring first to FIG. **2**, the substrate **10** with a mask layer **12** formed over substrate **10** are shown in accordance with some embodiments. The substrate **10** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like. The substrate **10** may be a wafer, such as a silicon wafer. FIG. **2** and the subsequent figures illustrate a portion of a wafer to better illustrate features of some embodiments. Similar structures and processes may be applied over larger portions of the wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a Buried Oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **10** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. In some embodiments, the mask layer **12** may include silicon oxide. The mask layer **12** may be formed by oxidizing a surface layer of the substrate **10** or any deposition methods such as using spin-on-coating, chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), atomic layer deposition (ALD), or the like. The mask layer **12** may include silicon oxide or other suitable materials. In some embodiments, the mask layer **12** protects or reduces the underlying substrate **10** from being damaged by subsequent implant processes. In some embodiments, the mask layer **12** can be omitted.

[0016] In some embodiments, the substrate **10** includes an n-type region **10**N, a p-type region **10**P, and an alignment mark region **10**A. The n-type region **10**N may be a region for fabricating n-type metal-oxide-semiconductor (NMOS) devices. The p-type region **10**P may be a region for fabricating p-type metal-oxide-semiconductor (PMOS) devices. The alignment mark region **10**A may be a region for fabricating alignment marks. The alignment marks that may be used to align the substrate **10** (e.g., a wafer) to a suitable direction or a suitable location, in the manufacturing processes.

[0017] A mask layer **14** having a pattern for alignment marks is formed over the mask layer **12**, in accordance with some embodiments. In accordance with some embodiments, the mask layer **14** may use spin-on-coating, chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), atomic layer deposition (ALD), physical vapor deposition (PVD), or the like. The mask layer **14** may include silicon nitride, silicon oxynitride, titanium nitride, silicon carbide, silicon oxide, a combination thereof, or the like. In some embodiments, the patterning of the mask layer **14** includes suitable photolithography and etch techniques.

[0018] Once the mask layer **14** has been formed, an etching process may be performed to etch the mask layer **12** and the substrate **10** through the mask layer **14**, thereby forming trenches **16** in the substrate **10** and the mask layer **12**, as shown in FIG. **3**A. In some embodiments, the depth D of the trenches **16** in the substrate is in a range from about 100 nm to about 150 nm, such as about 120 nm, beneath a major upper surface **10**S of the substrate **10**. Each trench **16** may have a width ranging from about 1 μm to about 2 μm, such as about 1.5 μm. The mask layer **14** may be removed after the trenches **16** are formed. In some embodiments, the mask layer **14** is removed by an etching process, such as a dry etch or a wet etch. The trenches and/or any materials formed therein may form an alignment mark. Although only 3 trenches **16** are illustrated in FIG. **3**A, the number and the arrangements of trenches **16** are not limited. For example, a top view of the alignment mark region **10**A of the substrate **10** is shown in FIG. **3**B, in accordance with some embodiments, and other shapes and arrangements of the trenches **16** may also be applied.

[0019] Referring to FIG. **4**, an implantation mask **18** is formed over the mask layer **12** and fills the trenches **16**, in accordance with some embodiments. The implantation mask **18** may be formed of a material capable of substantially blocking ions in an implantation process. For example, the implantation mask **18** may be a hard mask material of sufficient thickness, such as silicon nitride, silicon oxynitride, silicon carbide, a combination thereof, or the like. The implantation mask **18** may be formed by suitable deposition techniques, such as spin-on-coating, chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), physical vapor deposition (PVD), or the like.

[0020] Referring to FIG. **5**, the implantation mask **18** may be patterned to form openings **26** in the implantation mask **18**. The openings **26** may expose the n-type region **10**N and the alignment mark region **10**A. Patterning the implantation mask **18** may include suitable photolithography and etching techniques. For example, one or more photoresist layers (not shown) may be deposited over the implantation mask **18** and patterned by suitable photolithography techniques. An etching process is then performed on the implantation mask **18** through the patterned photoresist layer to form the openings **26** in the implantation mask **18**. An acceptable process, such as an ashing process, may be used to remove the photoresist layer.

[0021] Once the implantation mask **18** is formed, one or more p-type ion implantation processes **28** are performed to lightly dope the substrate **10**, thereby forming one or more p-wells **24** in the substrate **10**, in accordance with some embodiments. The p-wells **24** may provide active regions in the substrate **10** for fabricating NMOS devices. In some embodiments, the material of the implantation mask **18** in the trenches **16** and the substrate **10** in the alignment mark region **10**A are also doped with the p-type dopant. The p-type dopant may include boron, indium, the like, or combinations thereof. In some embodiments, the p-type ion implantation processes **28** include multiple blanket implantation processes to achieve the desired dopant profile. For example, a first p-type ion implantation may be performed using an energy in a range from about 180 keV to about 240 keV with a dose of about 1×10.sup.12 atoms/cm.sup.2 to about 1×10.sup.13 atoms/cm.sup.2. The first p-type implantation process may provide p-type regions in the substrate that may act as deep p-wells (not separately shown). A second p-type ion implantation may be performed using energy in a range from about 1 keV to about 100 keV with a dose of about 1×10.sup.13 atoms/cm.sup.2 to about 1×10.sup.14 atoms/cm.sup.2. The second p-type implantation process may provide p-type regions in the substrate that act as shallow p-wells (not separately shown). Although only two p-type ion implantation processes are described above as an example, more p-type ion implantations may be applied to achieve the desired dopant profile. The p-type dopant concentration in the p-well **24** and the material of the implantation mask **18** may be equal to or less than about 1×10.sup.16 cm.sup.−3, such as in a range from about 1×10.sup.15 cm.sup.−3 to about 5×10.sup.15 cm.sup.−3.

[0022] The first and second p-type implantation processes may be hot implantation processes, such as in a temperature of over about 150° C., such as of between about 200° C. and about 450° C. The hot implantation processes may help reduce interstitial defects generated in the substrate **10** during the implantation processes. Unlike a photoresist material, the hard mask material of the implantation mask **18** may not shrink or deform (or at least shrink and/or deform to a significantly lesser extent than does photoresist material) and can hence retain its pattern and critical dimensions after one or more hot implantation processes. The boundary of the p-well **24** in a lateral direction may be therefore precisely controlled with the use of implantation mask **18**. An annealing may be performed to repair implantation damage and to activate the implanted impurities. The annealing may be performed at a temperature in a range from about 1000° C. to about 1100° C., such as about 1050° C. for a duration of about 1 second to about 20 seconds, such as about 10 seconds.

[0023] Referring to FIG. **6**, once the p-well **24** has been formed, an etching process is performed to remove the implantation mask **18** over the major upper surface **10**S of the substrate **10**. In some embodiments, the etching process also partially etches the material of the implantation mask **18** in trenches **16**. The implantation mask **18** may not be completely etched because of loading effects. For example, the remaining material of the implantation mask **18** in the trenches **16** may have a thickness T.sub.1 of about 0.5 μm to about 1 μm. The etching process may be an acceptable etching process, such as a wet etch, a dry etch, or a combination thereof. The etching process may be a timed etching process, and the recessing depth or the thickness T.sub.1 may be controlled by the time of the etching process.
[0024] Referring to FIG. **7**, an implantation mask **30** is formed over the mask layer **12** and fills the trenches **16**, in accordance with some embodiments. For example, a portion of the material of the implantation mask **30** may be in the trenches **16** and above the material of the implantation mask **18**. The implantation mask **30** is formed of a material capable of substantially blocking ions in an implantation process. The implantation mask **30** may be a hard mask material of sufficient thickness, such as silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, a combination thereof, or the like. The implantation mask **30** may be a material different from the implantation mask **18** although a similar material may be used. The implantation mask **30** may be formed by suitable deposition techniques, such as spin-on-coating, chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), physical vapor deposition (PVD), or the like.
[0025] Referring to FIG. **8**, the implantation mask **30** is patterned to form openings **32** in the implantation mask **30**, in accordance with some embodiments. The openings **32** may expose the p-type region **10**P and the alignment mark region **10**A. Patterning the implantation mask **30** may include suitable photolithography and etching techniques. For example, one or more photoresist layers (not shown) may be deposited over the implantation mask **30** and patterned by suitable photolithography techniques. An etching process is then performed on the implantation mask **30** through the patterned photoresist layer to form the openings **32** in the implantation mask **30**. An acceptable process, such as an ashing process, may be used to remove the photoresist layer.
[0026] Once the implantation mask **30** has been formed, one or more n-type ion implantation processes **36** are performed to lightly dope the substrate **10**, thereby forming one or more n-wells **38** in the substrate **10**, in accordance with some embodiments. The n-wells **38** may provide active regions in the substrate **10** for fabricating PMOS devices. In some embodiments, the material of the implantation mask **30** in the trenches **16** and the substrate **10** in the alignment mark region **10**A are also doped with the n-type dopant. The n-type dopant may include phosphorous, arsenic, antimony, the like, or combinations thereof. In some embodiments, the n-type ion implantation processes **36** include multiple blanket implantation processes to achieve the desired dopant profile. For example, a first n-type ion implantation may be performed using energy in a range from about 180 keV to about 240 keV with a dose of about 1×10.sup.12 atoms/cm.sup.2 to about 1×10.sup.13 atoms/cm.sup.2. The first n-type implantation process may provide n-type regions in the substrate that may act as deep n-wells (not separately shown). A second n-type ion implantation may be performed using an energy in a range from about 1 keV to about 150 keV with a dose of 1×10.sup.13 atoms/cm.sup.2 to about 1×10.sup.14/cm.sup.2. The second n-type implantation process may provide n-type regions in the substrate that may act as shallow n-wells (not separately shown). Although only two n-type ion implantation processes are described above as an example, more n-type ion implantation may also be applied to achieve the desired dopant profile. The n-type dopant concentration in the n-well **38**, the material of implantation mask **18** in the trenches **16**, and the material of implantation mask **30** in the trenches **16** may be equal to or less than 1×10.sup.16 cm.sup.−3, such as in a range from about 1×10.sup.15 cm.sup.−3 to about 5×10.sup.15 cm.sup.−3. In some embodiments, the n-well **38** abuts the p-well **24**.
[0027] The first and second n-type implantation processes may be hot implantation processes, such as in a temperature of over about 150° C., such as of between about 200° C. and about 450° C. The

hot implantation processes may help reduce interstitial defects generated in the substrate **10** resulting from the implantation processes. Unlike a photoresist material, the hard mask material of the implantation mask **30** may not shrink or deform (or at least shrink and/or deform to a significantly lesser extent than does photoresist material) and hence can retain its pattern and critical dimensions after one or more hot implantation processes. The boundary of the n-well **38** in a lateral direction may be precisely controlled with the use of implantation mask **30**, which may help reduce overlapping of neighboring p-well **24** and n-well **38**. Junction leakage may be thus reduced. An annealing may be performed to repair implantation damage and to activate the implanted impurities. The annealing may be performed at a temperature in a range from about 1000° C. to about 1100° C., such as about 1050° C. for a duration of about 1 second to about 20 seconds, such as about 10 seconds.

[0028] Referring to FIG. **9**, once the n-well **38** has been formed, at least one etching process is performed to remove the implantation mask **30** over the major upper surface **10**S of the substrate **10**, and the mask layer **12**. For example, the implantation mask **30** and the mask layer **12** may be removed by separate etching process. Alternatively, the implantation mask **30** and the mask layer **12** may be removed together by a single etching process when the implantation mask **30** and the mask layer **12** are a similar material. In some embodiments, the material of implantation mask **30** in the trenches **16** are also partially or completely removed by the etching process. Any remaining material of the implantation mask **30** will be level with or below the major upper surface **10**S of the substrate **10**. The etching process may be an acceptable etching process, such as a wet etch, a dry etch, or a combination thereof. The etching process may be a timed etching process, and the recessing depth or the remaining thickness of the implantation mask **30** in the trenches **16** may be controlled by the time of the etching process.

[0029] The materials of the implantation mask **18** and the implantation mask **30** in the trenches **16** form an alignment mark **40** that may be used to align the wafer in the subsequent processes. A lower portion of the alignment mark **40** may be formed of the material of the implantation mask **18**, and an upper portion of the alignment mark **40** may be formed of the material of the implantation mask **30**. In some embodiments, the lower portion of the alignment mark **40** is doped with both p-type and n-type dopants, and the upper portion of the alignment mark **40** is doped with the n-type dopant. Alternatively, in some embodiments, the alignment mark **40** is doped with only one kind of dopant. For example, the openings **26** of the implantation mask **18** may not expose the alignment mark region **10**A, so the alignment mark **40** is doped with the n-type dopant only. Alternatively, the openings **32** of the implantation mask **30** may not expose the alignment mark region **10**A, so the lower portion of the alignment mark **40** is doped with the p-type dopant only, and the upper portion of the alignment mark **40** is non-doped. In some embodiments, the substrate **10** in the alignment mark region **10**A adjacent to the alignment mark **40** may also be doped with the same type and concentration of dopants as the alignment mark **40**. FIGS. **2** through **9** show forming the p-well **24** prior to forming the n-well **38** for illustrative purposes. In some embodiments, the n-well **38** is formed prior to the p-well **24**, and the conductivity types of the dopants in the alignment mark **40** are opposite to the embodiments as discussed above. For example, the lower portion of the alignment mark **40** may be doped with both the n-type dopant and p-type dopant or doped with only the n-type dopant, and the upper portion of alignment mark **40** may be doped with only the p-type dopant or non-doped.

[0030] Although FIGS. **2** to **9** illustrate the alignment mark **40** in the reference cross-section A-A′, the alignment mark **40** may be formed in other locations, such as in the cross-section B-B′ or C-C′ or in a cross-section parallel to the cross-section A-A′ or B-B′. In addition, although the alignment mark **40** is not illustrated in the nano-FET device illustrated in FIG. **1**, the alignment mark **40** may be in device regions of the substrate **10** and remain in the IC chips after the substrate **10** is scribed, or alternatively the alignment marks may be in scribe regions or other non-device regions of the substrate **10** and may not exist in the IC chips after the substrate **10** is scribed. For illustrative purposes, in the subsequent figures (e.g., starting from FIG. **10**), the alignment mark region **10**A is individually shown such as in FIGS. **11**B, **12**B, and **28**D and not separately shown in figures showing the reference cross-section A-A′.

[0031] Additional process steps may be performed to manufacture the nano-FETs. Referring to FIG. **10**, a multi-layer stack **64** is formed over the substrate **10**. The multi-layer stack **64** includes alternating layers of first semiconductor layers **51**A-C (collectively referred to as first semiconductor layers **51**) and second semiconductor layers **53**A-C (collectively referred to as second semiconductor layers **53**). For purposes of illustration and as discussed in greater detail below, the second semiconductor layers **53** will be removed and the first semiconductor layers **51** will be patterned to form channel regions of nano-FETs in the p-type region **10**P. Also, the first semiconductor layers **51** will be removed and the second semiconductor layers **53** will be patterned to form channel regions of nano-FETs in the n-type region **10**N. Nevertheless, in some embodiments the first semiconductor layers **51** may be removed and the second semiconductor layers **53** may be patterned to form channel regions of nano-FETs in the n-type region **10**N, and the second semiconductor layers **53** may be removed and the first semiconductor layers **51** may be patterned to form channel regions of nano-FETs in the p-type region **10**P.

[0032] In some embodiments, the first semiconductor layers **51** may be removed and the second semiconductor layers **53** may be patterned to form channel regions of nano-FETS in both the n-type region **10**N and the p-type region **10**P. In other embodiments, the second semiconductor layers **53** may be removed and the first semiconductor layers **51** may be patterned to form channel regions of non-FETs in both the n-type region **10**N and the p-type region **10**P. In such embodiments, the channel regions in both the n-type region **10**N and the p-type region **10**P may have a same material composition (e.g., silicon, or the another semiconductor material) and be formed simultaneously. FIGS. **29**A-C illustrate a structure resulting from such embodiments where the channel regions in both the p-type region **10**P and the n-type region **10**N comprise silicon, for example.

[0033] The multi-layer stack **64** is illustrated as including three layers of each of the first semiconductor layers **51** and the second semiconductor layers **53** for illustrative purposes. In some embodiments, the multi-layer stack **64** may include any number of the first semiconductor layers **51** and the second semiconductor layers **53**. Each of the layers of the multi-layer stack **64** may be epitaxially grown using a process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), vapor phase epitaxy (VPE), molecular beam epitaxy (MBE), or the like. In various embodiments, the first semiconductor layers **51** may be formed of a first semiconductor material suitable for p-type nano-FETs, such as silicon germanium, or the like, and the second semiconductor layers **53** may be formed of a second semiconductor material suitable for n-type nano-FETs, such as silicon, silicon carbon, or the like. The multi-layer stack **64** is illustrated as having a bottommost semiconductor layer suitable for p-type nano-FETs for illustrative purposes. In some embodiments, multi-layer stack **64** may be formed such that the bottommost layer is a semiconductor layer suitable for n-type nano-FETs. The first semiconductor layers **51** and the second semiconductor layers **53** may be doped in situ or using one or more implant processes.

[0034] The first semiconductor materials and the second semiconductor materials may be materials having a high-etch selectivity to one another. As such, the first semiconductor layers **51** of the first semiconductor material may be removed without significantly removing the second semiconductor layers **53** of the second semiconductor material in the n-type region **10**N, thereby allowing the second semiconductor layers **53** to be patterned to form channel regions of n-type nano-FETs. Similarly, the second semiconductor layers **53** of the second semiconductor material may be removed without significantly removing the first semiconductor layers **51** of the first semiconductor material in the p-type region **10**P, thereby allowing the first semiconductor layers **51** to be patterned to form channel regions of p-type nano-FETs.

[0035] Referring now to FIG. **11**A, fins **66** are formed in the regions of substrate **10** and nanostructures **55** are formed in the multi-layer stack **64**, in accordance with some embodiments.

The fins **66** protrude from a major upper surface **10**S of the substrate **10** and the height of the fins **66** is in a range from about 50 nm to about 70 nm. In some embodiments, the nanostructures **55** and the fins **66** may be formed in the multi-layer stack **64** and the substrate **10**, respectively, by etching trenches in the multi-layer stack **64** and the substrate **10**. In some embodiments, referring to FIG. **11**B, the alignment mark **40** may be also recessed with the substrate **10** while forming the fins **66** in the p-type region **10**P and the n-type region **10**B. For example, at least a portion of the material of the implantation mask **30** is etched. Accordingly, the top surface of the alignment mark **40** may be below or even with the major upper surface **10**S of the substrate **10**. The remaining material of the implantation mask **30** in the trenches **16** may have a thickness T.sub.2 thinner than the thickness T.sub.1 of the material of the implantation mask **18** in the trenches **16**. For example, the thickness T.sub.2 may be smaller than about 0.5 μm.
[0036] The etching process to form the fins **66** and the nanostructures **55** may be any acceptable etching process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. Forming the nanostructures **55** by etching the multi-layer stack **64** may further define first nanostructures **52**A-C (collectively referred to as the first nanostructures **52**) from the first semiconductor layers **51** and define second nanostructures **54**A-C (collectively referred to as the second nanostructures **54**) from the second semiconductor layers **53**. The first nanostructures **52** and the second nanostructures **54** may further be collectively referred to as nanostructures **55**. FIG. **11**A illustrates that two fins are formed in each of the n-type region **10**N and the p-type region **10**P, in other embodiments a different number of fins may be formed in each region.
[0037] The fins **66** and the nanostructures **55** may be patterned by any suitable method. For example, the fins **66** and the nanostructures **55** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins **66**.
[0038] FIG. **11**A illustrates the fins **66** in the n-type region **10**N and the p-type region **10**P as having substantially equal widths for illustrative purposes. In some embodiments, widths of the fins **66** in the n-type region **10**N may be greater or thinner than the fins **66** in the p-type region **10**P. Further, while each of the fins **66** and the nanostructures **55** are illustrated as having a consistent width throughout, in other embodiments, the fins **66** and/or the nanostructures **55** may have tapered sidewalls such that a width of each of the fins **66** and/or the nanostructures **55** continuously increases in a direction towards the substrate **10**. In such embodiments, each of the nanostructures **55** may have a different width and be trapezoidal in shape.
[0039] In FIGS. **12**A and **12**B, shallow trench isolation (STI) regions **68** are formed adjacent the fins **66** and above the alignment mark **40**, in accordance with some embodiments. The STI regions **68** may be formed by depositing an insulation material over the substrate **10**, the fins **66**, and nanostructures **55**, between adjacent fins **66**, and above the alignment mark **40**. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by high-density plasma CVD (HDP-CVD), flowable CVD (FCVD), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material is silicon oxide formed by an FCVD process. An annealing process may be performed once the insulation material is formed. In an embodiment, the insulation material is formed such that excess insulation material covers the nanostructures **55**. Although the insulation material is illustrated as a single layer, some embodiments may utilize

multiple layers. For example, in some embodiments a liner (not separately illustrated) may first be formed along the major upper surface **10**S of the substrate **10**, the fins **66**, and the nanostructures **55**. Thereafter, a fill material, such as those discussed above may be formed over the liner.

[0040] A removal process is then applied to the insulation material to remove excess insulation material over the nanostructures **55**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process exposes the nanostructures **55** such that top surfaces of the nanostructures **55** and the insulation material are level after the planarization process is complete.

[0041] The insulation material is then recessed to form the STI regions **68**. The insulation material is recessed such that upper portions of fins **66** in the n-type region **10**N and the p-type region **10**P protrude from between neighboring STI regions **68**. Further, the top surfaces of the STI regions **68** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **68** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **68** may be recessed using an acceptable etching process, such as one that is selective to the material of the insulation material (e.g., etches the material of the insulation material at a faster rate than the material of the fins **66** and the nanostructures **55**). For example, an oxide removal using, for example, dilute hydrofluoric (dHF) acid may be used.

[0042] In some embodiments, referring to FIG. **12**B, the STI regions **68** includes a portion filling the trenches **16** and covering the alignment mark **40**. The alignment mark **40** will be covered by the STI regions **68** and not substantially altered in subsequent processes. In some embodiments, the material of the implantation mask **30** and the STI regions **68** are both formed of oxide, and the implantation mask **30** has a higher density than the STI regions **68** when the implantation mask **30** is formed by PECVD, and the STI regions **68** is formed by FCVD.

[0043] The process described above with respect to FIGS. **10**, **11**A, and **12**A is just one example of how the fins **66** and the nanostructures **55** may be formed. In some embodiments, the fins **66** and/or the nanostructures **55** may be formed using a mask and an epitaxial growth process. For example, a dielectric layer can be formed over the major upper surface **10**S of the substrate **10**, and trenches can be etched through the dielectric layer to expose the underlying substrate **10**. Epitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the epitaxial structures protrude from the dielectric layer to form the fins **66** and/or the nanostructures **55**. The epitaxial structures may comprise the alternating semiconductor materials discussed above, such as the first semiconductor materials and the second semiconductor materials. In some embodiments, where epitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, or doped through one or more implantation processes.

[0044] Additionally, the first semiconductor layers **51** (and resulting nanostructures **52**) and the second semiconductor layers **53** (and resulting nanostructures **54**) are illustrated and discussed herein as comprising the same materials in the p-type region **10**P and the n-type region **10**N for illustrative purposes only. As such, in some embodiments one or both of the first semiconductor layers **51** and the second semiconductor layers **53** may be different materials or formed in a different order in the p-type region **10**P and the n-type region **10**N.

[0045] In FIG. **13**, a dummy dielectric layer **70** is formed on the fins **66** and/or the nanostructures **55**. The dummy dielectric layer **70** may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **72** is formed over the dummy dielectric layer **70**, and a mask layer **74** is formed over the dummy gate layer **72**. The dummy gate layer **72** may be deposited over the dummy dielectric layer **70** and then planarized, such as by a CMP. The mask layer **74** may be deposited over the dummy gate layer **72**. The dummy gate layer **72** may be a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer **72** may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques for depositing the selected material. The dummy gate layer **72** may be made of other materials that have a high etching selectivity from the etching of isolation regions. The mask layer **74** may include, for example, silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer **72** and a single mask layer **74** are formed across the n-type region **10**N and the p-type region **10**P. It is noted that the dummy dielectric layer **70** is shown covering only the fins **66** and the nanostructures **55** for illustrative purposes only. In some embodiments, the dummy dielectric layer **70** may be deposited such that the dummy dielectric layer **70** covers the STI regions **68**, such that the dummy dielectric layer **70** extends between the dummy gate layer **72** and the STI regions **68**.

[0046] FIGS. **14**A through **26**C illustrate various additional steps in the manufacturing of embodiment devices. In FIGS. **14**A and **14**B, the mask layer **74** (see FIG. **13**) may be patterned using acceptable photolithography and etching techniques to form masks **78**. The pattern of the masks **78** then may be transferred to the dummy gate layer **72** and to the dummy dielectric layer **70** to form dummy gates **76** and dummy gate dielectrics **71**, respectively. The dummy gates **76** cover respective channel regions of the fins **66**. The pattern of the masks **78** may be used to physically separate each of the dummy gates **76** from adjacent dummy gates **76**. The dummy gates **76** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective fins **66**.

[0047] In FIGS. **15**A and **15**B, a first spacer layer **80** and a second spacer layer **82** are formed over the structures illustrated in FIGS. **14**A and **14**B, respectively. The first spacer layer **80** and the second spacer layer **82** will be subsequently patterned to act as spacers for forming self-aligned source/drain regions. In FIGS. **15**A and **15**B, the first spacer layer **80** is formed on top surfaces of the STI regions **68**; top surfaces and sidewalls of the fins **66**, the nanostructures **55**, and the masks **78**; and sidewalls of the dummy gates **76** and the dummy gate dielectric **71**. The second spacer layer **82** is deposited over the first spacer layer **80**. The first spacer layer **80** may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like, using techniques such as thermal oxidation or deposited by CVD, ALD, or the like. The second spacer layer **82** may be formed of a material having a different etch rate than the material of the first spacer layer **80**, such as silicon oxide, silicon nitride, silicon oxynitride, or the like, and may be deposited by CVD, ALD, or the like.

[0048] After the first spacer layer **80** is formed and prior to forming the second spacer layer **82**, implantations for lightly doped source/drain (LDD) regions (not separately illustrated) may be performed. In embodiments with different device types, a mask, such as a photoresist, may be formed over the n-type region **10**N, while exposing the p-type region **10**P, and appropriate type (e.g., p-type) impurities may be implanted into the exposed fins **66** and nanostructures **55** in the p-type region **10**P. The mask may then be removed. Subsequently, a mask, such as a photoresist, may be formed over the p-type region **10**P while exposing the n-type region **10**N, and appropriate type impurities (e.g., n-type) may be implanted into the exposed fins **66** and nanostructures **55** in the n-type region **10**N. The mask may then be removed. The n-type impurities may be the any of the n-type impurities previously discussed, and the p-type impurities may be the any of the p-type impurities previously discussed. The lightly doped source/drain regions may have a concentration of impurities in a range from about 1×10.sup.15 atom/cm.sup.3 to about 1×10.sup.19 atom/cm.sup.3. An annealing may be used to repair implantation damage and to activate the implanted impurities.

[0049] In FIGS. **16**A and **16**B, the first spacer layer **80** and the second spacer layer **82** are etched to form first spacers **81** and second spacers **83**. As will be discussed in greater detail below, the first spacers **81** and the second spacers **83** act to self-aligned subsequently formed source drain regions, as well as to protect sidewalls of the fins **66** and/or nanostructure **55** during subsequent processing. The first spacer layer **80** and the second spacer layer **82** may be etched using a suitable etching process, such as an isotropic etching process (e.g., a wet etching process), an anisotropic etching process (e.g., a dry etching process), or the like. In some embodiments, the material of the second spacer layer **82** has a different etch rate than the material of the first spacer layer **80**, such that the first spacer layer **80** may act as an etch stop layer when patterning the second spacer layer **82** and such that the second spacer layer **82** may act as a mask when patterning the first spacer layer **80**. For example, the second spacer layer **82** may be etched using an anisotropic etching process wherein the first spacer layer **80** acts as an etch stop layer, wherein remaining portions of the second spacer layer **82** form second spacers **83** as illustrated in FIG. **16**A. Thereafter, the second spacers **83** acts as a mask while etching exposed portions of the first spacer layer **80**, thereby forming first spacers **81** as illustrated in FIG. **16**A.

[0050] As illustrated in FIG. **16**A, the first spacers **81** and the second spacers **83** are disposed on sidewalls of the fins **66** and/or nanostructures **55**. As illustrated in FIG. **8**B, in some embodiments, the second spacer layer **82** may be removed from over the first spacer layer **80** adjacent the masks **78**, the dummy gates **76**, and the dummy gate dielectrics **71**, and the first spacers **81** are disposed on sidewalls of the masks **78**, the dummy gates **76**, and the dummy dielectric layers **60**. In other embodiments, a portion of the second spacer layer **82** may remain over the first spacer layer **80** adjacent the masks **78**, the dummy gates **76**, and the dummy gate dielectrics **71**.

[0051] It is noted that the above disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, different sequence of steps may be utilized (e.g., the first spacers **81** may be patterned prior to depositing the second spacer layer **82**), additional spacers may be formed and removed, and/or the like. Furthermore, the n-type and p-type devices may be formed using different structures and steps.

[0052] In FIGS. **17**A and **17**B, first recesses **86** are formed in the fins **66**, the nanostructures **55**, and the substrate **10**, in accordance with some embodiments. Epitaxial source/drain regions will be subsequently formed in the first recesses **86**. The first recesses **86** may extend through the first nanostructures **52** and the second nanostructures **54**, and into the substrate **10**. As illustrated in FIG. **17**A, top surfaces of the STI regions **68** may be level with bottom surfaces of the first recesses **86**. In various embodiments, the fins **66** may be etched such that bottom surfaces of the first recesses **86** are disposed below the top surfaces of the STI regions **68**; or the like. The first recesses **86** may be formed by etching the fins **66**, the nanostructures **55**, and the substrate **10** using anisotropic etching processes, such as RIE, NBE, or the like. The first spacers **81**, the second spacers **83**, and the masks **78** mask portions of the fins **66**, the nanostructures **55**, and the substrate **10** during the etching processes used to form the first recesses **86**. A single etching process or multiple etching processes may be used to etch each layer of the nanostructures **55** and/or the fins **66**. Timed etching processes may be used to stop the etching of the first recesses **86** after the first recesses **86** reach a desired depth.

[0053] In FIGS. **18**A and **18**B, portions of sidewalls of the layers of the multi-layer stack **64** formed of the first semiconductor materials (e.g., the first nanostructures **52**) exposed by the first recesses **86** are etched to form sidewall recesses **88** in the n-type region **10**N, and portions of sidewalls of the layers of the multi-layer stack **56** formed of the second semiconductor materials (e.g., the second nanostructures **54**) exposed by the first recesses **86** are etched to form sidewall recesses **88** in the p-type region **10**P. Although sidewalls of the first nanostructures **52** and the second nanostructures **54** in sidewall recesses **88** are illustrated as being straight in FIG. **20**B, the sidewalls may be concave or convex. The sidewalls may be etched using isotropic etching processes, such as wet etching or the like. The p-type region **10**P may be protected using a mask (not shown) while etchants selective to the first semiconductor materials are used to etch the first nanostructures **52** such that the second nanostructures **54** and the substrate **10** remain relatively unetched as compared to the first nanostructures **52** in the n-type region **10**N. Similarly, the n-type region **10**N may be protected using a mask (not shown) while etchants selective to the second semiconductor materials are used to etch the second nanostructures **54** such that the first nanostructures **52** and the substrate **10** remain relatively unetched as compared to the second nanostructures **54** in the p-type region **10**P. In an embodiment in which the first nanostructures **52** include, e.g., SiGe, and the second nanostructures **54** include, e.g., Si or SiC, a dry etching process with tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH.sub.4OH), or the like may be used to etch sidewalls of the first nanostructures **52** in the n-type region **10**N, and a wet or dry etching process with hydrogen fluoride, another fluorine-based etchant, or the like may be used to etch sidewalls of the second nanostructures **54** in the p-type region **10**P.

[0054] In FIGS. **19**A-**19**C, first inner spacers **90** are formed in the sidewall recess **88**. The first inner spacers **90** may be formed by depositing an inner spacer layer (not separately illustrated) over the structures illustrated in FIGS. **20**A and **20**B. The first inner spacers **90** act as isolation features between subsequently formed source/drain regions and a gate structure. As will be discussed in greater detail below, source/drain regions will be formed in the first recesses **86**, while the first nanostructures **52** in the n-type region **10**N and the second nanostructures **54** in the p-type region **10**P will be replaced with corresponding gate structures.

[0055] The inner spacer layer may be deposited by a conformal deposition process, such as CVD, ALD, or the like. The inner spacer layer may comprise a material such as silicon nitride or silicon oxynitride, although any suitable material, such as low-dielectric constant (low-k) materials having a k-value less than about 3.5, may be utilized. The inner spacer layer may then be anisotropically etched to form the first inner spacers **90**. Although outer sidewalls of the first inner spacers **90** are illustrated as being flush with sidewalls of the second nanostructures **54** in the n-type region **10**N and flush with the sidewalls of the first nanostructures **52** in the p-type region **10**P, the outer sidewalls of the first inner spacers **90** may extend beyond or be recessed from sidewalls of the second nanostructures **54** and/or the first nanostructures **52**, respectively.

[0056] Moreover, although the outer sidewalls of the first inner spacers **90** are illustrated as being straight in FIG. **19**B, the outer sidewalls of the first inner spacers **90** may be concave or convex. As an example, FIG. **19**C illustrates an embodiment in which sidewalls of the first nanostructures **52** are concave, outer sidewalls of the first inner spacers **90** are concave, and the first inner spacers **90** are recessed from sidewalls of the second nanostructures **54** in the n-type region **10**N. Also illustrated are embodiments in which sidewalls of the second nanostructures **54** are concave, outer sidewalls of the first inner spacers **90** are concave, and the first inner spacers **90** are recessed from sidewalls of the first nanostructures **52** in the p-type region **10**P. The inner spacer layer may be etched by an anisotropic etching process, such as RIE, NBE, or the like. The first inner spacers **90** may be used to prevent damage to subsequently formed source/drain regions (such as the epitaxial source/drain regions **92**, discussed below with respect to FIGS. **20**A-**20**C) by subsequent etching processes, such as etching processes used to form gate structures.

[0057] In FIGS. **20**A-**20**C, epitaxial source/drain regions **92** are formed in the first recesses **86**. In some embodiments, the source/drain regions **92** may exert stress on the second nanostructures **54** in the n-type region **10**N and on the first nanostructures **52** in the p-type region **10**P, thereby improving performance. As illustrated in FIG. **20**B, the epitaxial source/drain regions **92** are formed in the first recesses **86** such that each dummy gate **76** is disposed between respective neighboring pairs of the epitaxial source/drain regions **92**. In some embodiments, the first spacers **81** are used to separate the epitaxial source/drain regions **92** from the dummy gates **76** and the first inner spacers **90** are used to separate the epitaxial source/drain regions **92** from the nanostructures **55** by an appropriate lateral distance so that the epitaxial source/drain regions **92** do not short out with subsequently formed gates of the resulting nano-FETs.

[0058] The epitaxial source/drain regions **92** in the n-type region **10**N, e.g., the NMOS region, may be formed by masking the p-type region **10**P, e.g., the PMOS region. Then, the epitaxial source/drain regions **92** are epitaxially grown in the first recesses **86** in the n-type region **10**N. The epitaxial source/drain regions **92** may include any acceptable material appropriate for n-type nano-

FETs. For example, if the second nanostructures **54** are silicon, the epitaxial source/drain regions **92** may include materials exerting a tensile strain on the second nanostructures **54**, such as silicon, silicon carbide, phosphorous doped silicon carbide, silicon phosphide, or the like. The epitaxial source/drain regions **92** may have surfaces raised from respective upper surfaces of the nanostructures **55** and may have facets.

[0059] The epitaxial source/drain regions **92** in the p-type region **10**P, e.g., the PMOS region, may be formed by masking the n-type region **10**N, e.g., the NMOS region. Then, the epitaxial source/drain regions **92** are epitaxially grown in the first recesses **86** in the p-type region **10**P. The epitaxial source/drain regions **92** may include any acceptable material appropriate for p-type nano-FETs. For example, if the first nanostructures **52** are silicon germanium, the epitaxial source/drain regions **92** may comprise materials exerting a compressive strain on the first nanostructures **52**, such as silicon-germanium, boron doped silicon-germanium, germanium, germanium tin, or the like. The epitaxial source/drain regions **92** may also have surfaces raised from respective surfaces of the multi-layer stack **56** and may have facets.

[0060] The epitaxial source/drain regions **92**, the first nanostructures **52**, the second nanostructures **54**, and/or the substrate **10** may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an annealing. The source/drain regions may have an impurity concentration of between about 1×10.sup.19 atom/cm.sup.3 and about 1×10.sup.21 atom/cm.sup.3. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the epitaxial source/drain regions **92** may be in situ doped during growth.

[0061] As a result of the epitaxy processes used to form the epitaxial source/drain regions **92** in the n-type region **10**N and the p-type region **10**P, upper surfaces of the epitaxial source/drain regions **92** have facets which expand laterally outward beyond sidewalls of the nanostructures **55**. In some embodiments, these facets cause adjacent epitaxial source/drain regions **92** of a same nano-FET to merge as illustrated by FIG. **20**A. In other embodiments, adjacent epitaxial source/drain regions **92** remain separated after the epitaxy process is completed as illustrated by FIG. **20**C. In the embodiments illustrated in FIGS. **20**A and **20**C, the first spacers **81** may be formed to a top surface of the STI regions **68** thereby blocking the epitaxial growth. In some other embodiments, the first spacers **81** may cover portions of the sidewalls of the nanostructures **55** further blocking the epitaxial growth. In some other embodiments, the spacer etch used to form the first spacers **81** may be adjusted to remove the spacer material to allow the epitaxially grown region to extend to the surface of the STI regions **68**.

[0062] The epitaxial source/drain regions **92** may comprise one or more semiconductor material layers. For example, the epitaxial source/drain regions **92** may comprise a first semiconductor material layer **92**A, a second semiconductor material layer **92**B, and a third semiconductor material layer **92**C. Any number of semiconductor material layers may be used for the epitaxial source/drain regions **92**. Each of the first semiconductor material layer **92**A, the second semiconductor material layer **92**B, and the third semiconductor material layer **92**C may be formed of different semiconductor materials and may be doped to different dopant concentrations. In some embodiments, the first semiconductor material layer **92**A may have a dopant concentration less than the second semiconductor material layer **92**B and greater than the third semiconductor material layer **92**C. In embodiments in which the epitaxial source/drain regions **92** comprise three semiconductor material layers, the first semiconductor material layer **92**A may be deposited, the second semiconductor material layer **92**B may be deposited over the first semiconductor material layer **92**A, and the third semiconductor material layer **92**C may be deposited over the second semiconductor material layer **92**B.

[0063] FIG. **20**D illustrates an embodiment in which sidewalls of the first nanostructures **52** in the n-type region **10**N and sidewalls of the second nanostructures **54** in the p-type region **10**P are concave, outer sidewalls of the first inner spacers **90** are concave, and the first inner spacers **90** are recessed from sidewalls of the second nanostructures **54** and the first nanostructures **52**, respectively. As illustrated in FIG. **20**D, the epitaxial source/drain regions **92** may be formed in contact with the first inner spacers **90** and may extend past sidewalls of the second nanostructures **54** in the n-type region **10**N and past sidewalls of the first nanostructures **52** in the p-type region **10**P.

[0064] In FIGS. **21**A-**21**C, a first interlayer dielectric (ILD) **96** is deposited over the structure illustrated in FIGS. **14**A, **20**B, and **20**A (the processes of FIGS. **15**A-**20**D do not alter the cross-section illustrated in FIG. **14**A), respectively. The first ILD **96** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) **94** is disposed between the first ILD **96** and the epitaxial source/drain regions **92**, the masks **78**, and the first spacers **81**. The CESL **94** may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a different etch rate than the material of the overlying first ILD **96**.

[0065] In FIGS. **22**A-**22**C, a planarization process, such as a CMP, may be performed to level the top surface of the first ILD **96** with the top surfaces of the dummy gates **76** or the masks **78**. The planarization process may also remove the masks **78** on the dummy gates **76**, and portions of the first spacers **81** along sidewalls of the masks **78**. After the planarization process, top surfaces of the dummy gates **76**, the first spacers **81**, and the first ILD **96** are level within process variations. Accordingly, the top surfaces of the dummy gates **72** are exposed through the first ILD **96**. In some embodiments, the masks **78** may remain, in which case the planarization process levels the top surface of the first ILD **96** with top surface of the masks **78** and the first spacers **81**.

[0066] In FIGS. **23**A and **23**B, the dummy gates **76**, and the masks **78** if present, are removed in one or more etching steps, so that second recesses **98** are formed. Portions of the dummy dielectric layers **60** in the second recesses **98** are also be removed. In some embodiments, the dummy gates **76** and the dummy dielectric layers **60** are removed by an anisotropic dry etching process. For example, the etching process may include a dry etching process using reaction gas(es) that selectively etch the dummy gates **76** at a faster rate than the first ILD **96** or the first spacers **81**. Each second recess **98** exposes and/or overlies portions of nanostructures **55**, which act as channel regions in subsequently completed nano-FETs. Portions of the nanostructures **55** which act as the channel regions are disposed between neighboring pairs of the epitaxial source/drain regions **92**. During the removal, the dummy dielectric layers **60** may be used as etch stop layers when the dummy gates **76** are etched. The dummy dielectric layers **60** may then be removed after the removal of the dummy gates **76**.

[0067] In FIGS. **24**A and **24**B, the first nanostructures **52** in the n-type region **10**N and the second nanostructures **54** in the p-type region **10**P are removed extending the second recesses **98**. The first nanostructures **52** may be removed by forming a mask (not shown) over the p-type region **10**P and performing an isotropic etching process such as wet etching or the like using etchants which are selective to the materials of the first nanostructures **52**, while the second nanostructures **54**, the substrate **10**, the STI regions **68** remain relatively unetched as compared to the first nanostructures **52**. In embodiments in which the first nanostructures **52** include, e.g., SiGe, and the second nanostructures **54**A-**54**C include, e.g., Si or SiC, tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH.sub.4OH), or the like may be used to remove the first nanostructures **52** in the n-type region **10**N.

[0068] The second nanostructures **54** in the p-type region **10**P may be removed by forming a mask (not shown) over the n-type region **10**N and performing an isotropic etching process such as wet etching or the like using etchants which are selective to the materials of the second nanostructures **54**, while the first nanostructures **52**, the substrate **10**, the STI regions **68** remain relatively unetched as compared to the second nanostructures **54**. In embodiments in which the second nanostructures **54** include, e.g., SiGe, and the first nanostructures **52** include, e.g., Si or SiC, hydrogen fluoride, another fluorine-based etchant, or the like may be used to remove the second nanostructures **54** in the p-type region **10**P.

[0069] In other embodiments, the channel regions in the n-type region **10**N and the p-type region **10**P may be formed simultaneously, for example by removing the first nanostructures **52** in both the n-type region **10**N and the p-type region **10**P or by removing the second nanostructures **54** in both the n-type region **10**N and the p-type region **10**P. In such embodiments, channel regions of n-type nano-FETs and p-type nano-FETS may have a same material composition, such as silicon, silicon germanium, or the like. FIGS. **29**A, **29**B, and **29**C illustrate a structure resulting from such embodiments where the channel regions in both the p-type region **10**P and the n-type region **10**N are provided by the second nanostructures **54** and comprise silicon, for example.

[0070] In FIGS. **25**A and **25**B, gate dielectric layers **100** and gate electrodes **102** are formed for replacement gates. The gate dielectric layers **100** are deposited conformally in the second recesses **98**. In the n-type region **10**N, the gate dielectric layers **100** may be formed on top surfaces and sidewalls of the substrate **10** and on top surfaces, sidewalls, and bottom surfaces of the second nanostructures **54**, and in the p-type region **10**P, the gate dielectric layers **100** may be formed on top surfaces and sidewalls of the substrate **10** and on top surfaces, sidewalls, and bottom surfaces of the first nanostructures **52**. The gate dielectric layers **100** may also be deposited on top surfaces of the first ILD **96**, the CESL **94**, the first spacers **81**, and the STI regions **68**.

[0071] In accordance with some embodiments, the gate dielectric layers **100** comprise one or more dielectric layers, such as an oxide, a metal oxide, the like, or combinations thereof. For example, in some embodiments, the gate dielectrics may comprise a silicon oxide layer and a metal oxide layer over the silicon oxide layer. In some embodiments, the gate dielectric layers **100** include a high-k dielectric material, and in these embodiments, the gate dielectric layers **100** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The structure of the gate dielectric layers **100** may be the same or different in the n-type region **10**N and the p-type region **10**P. The formation methods of the gate dielectric layers **100** may include molecular-beam deposition (MBD), ALD, PECVD, and the like.

[0072] The gate electrodes **102** are deposited over the gate dielectric layers **100**, respectively, and fill the remaining portions of the second recesses **98**. The gate electrodes **102** may include a metal-containing material such as titanium nitride, titanium oxide, tantalum nitride, tantalum carbide, cobalt, ruthenium, aluminum, tungsten, combinations thereof, or multi-layers thereof. For example, although single layer gate electrodes **102** are illustrated in FIGS. **25**A and **25**B, the gate electrodes **102** may comprise any number of liner layers, any number of work function tuning layers, and a fill material. Any combination of the layers which make up the gate electrodes **102** may be deposited in the n-type region **10**N between adjacent ones of the second nanostructures **54** and between the second nanostructure **54**A and the substrate **10**, and may be deposited in the p-type region **10**P between adjacent ones of the first nanostructures **52**.

[0073] The formation of the gate dielectric layers **100** in the n-type region **10**N and the p-type region **10**P may occur simultaneously such that the gate dielectric layers **100** in each region are formed from the same materials, and the formation of the gate electrodes **102** may occur simultaneously such that the gate electrodes **102** in each region are formed from the same materials. In some embodiments, the gate dielectric layers **100** in each region may be formed by distinct processes, such that the gate dielectric layers **100** may be different materials and/or have a different number of layers, and/or the gate electrodes **102** in each region may be formed by distinct processes, such that the gate electrodes **102** may be different materials and/or have a different number of layers. Various masking steps may be used to mask and expose appropriate regions when using distinct processes.

[0074] After the filling of the second recesses **98**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectric layers **100** and the material of the gate electrodes **102**, which excess portions are over the top surface of the first ILD **96**. The remaining portions of material of the gate electrodes **102** and the gate dielectric layers **100** thus form replacement gate structures of the resulting nano-FETs. The gate electrodes **102** and the gate dielectric layers **100** may be collectively referred to as “gate structures.”

[0075] In FIGS. **26**A-**26**C, the gate structure (including the gate dielectric layers **100** and the corresponding overlying gate electrodes **102**) is recessed, so that a recess is formed directly over the gate structure and between opposing portions of first spacers **81**. A gate mask **104** comprising one or more layers of dielectric material, such as silicon nitride, silicon oxynitride, or the like, is filled in the recess, followed by a planarization process to remove excess portions of the dielectric material extending over the first ILD **96**. Subsequently formed gate contacts (such as the contacts **114**, discussed below with respect to FIGS. **28**A and **28**B) penetrate through the gate mask **104** to contact the top surface of the recessed gate electrodes **102**.

[0076] As further illustrated by FIGS. **26**A-**26**C, a second ILD **106** is deposited over the first ILD **96** and over the gate mask **104**. In some embodiments, the second ILD **106** is a flowable film formed by FCVD. In some embodiments, the second ILD **106** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD, PECVD, or the like.

[0077] In FIGS. **27**A-**27**C, the second ILD **106**, the first ILD **96**, the CESL **94**, and the gate masks **104** are etched to form third recesses **108** exposing surfaces of the epitaxial source/drain regions **92** and/or the gate structure. The third recesses **108** may be formed by etching using an anisotropic etching process, such as RIE, NBE, or the like. In some embodiments, the third recesses **108** may be etched through the second ILD **106** and the first ILD **96** using a first etching process; may be etched through the gate masks **104** using a second etching process; and may then be etched through the CESL **94** using a third etching process. A mask, such as a photoresist, may be formed and patterned over the second ILD **106** to mask portions of the second ILD **106** from the first etching process and the second etching process. In some embodiments, the etching process may over-etch, and therefore, the third recesses **108** extend into the epitaxial source/drain regions **92** and/or the gate structure, and a bottom of the third recesses **108** may be level with (e.g., at a same level, or having a same distance from the substrate), or lower than (e.g., closer to the substrate) the epitaxial source/drain regions **92** and/or the gate structure. Although FIG. **29**B illustrate the third recesses **108** as exposing the epitaxial source/drain regions **92** and the gate structure in a same cross section, in various embodiments, the epitaxial source/drain regions **92** and the gate structure may be exposed in different cross-sections, thereby reducing the risk of shorting subsequently formed contacts. After the third recesses **108** are formed, silicide regions **110** are formed over the epitaxial source/drain regions **92**. In some embodiments, the silicide regions **110** are formed by first depositing a metal (not shown) capable of reacting with the semiconductor materials of the underlying epitaxial source/drain regions **92** (e.g., silicon, silicon germanium, germanium) to form silicide or germanide regions, such as nickel, cobalt, titanium, tantalum, platinum, tungsten, other noble metals, other refractory metals, rare earth metals or their alloys, over the exposed portions of the epitaxial source/drain regions **92**, then performing a thermal annealing process to form the silicide regions **110**. The un-reacted portions of the deposited metal are then removed, e.g., by an etching process. Although silicide regions **110** are referred to as silicide regions, silicide regions **110** may also be germanide regions, or silicon germanide regions (e.g., regions comprising silicide and germanide). In an embodiment, the silicide region **110** comprises TiSi, and has a thickness in a range between about 2 nm and about 10 nm.

[0078] Next, in FIGS. **28**A-C, contacts **112** and **114** (may also be referred to as contact plugs) are formed in the third recesses **108**. The contacts **112** and **114** may each comprise one or more layers, such as barrier layers, diffusion layers, and fill materials. For example, in some embodiments, the contacts **112** and **114** each include a barrier layer and a conductive material, and is electrically coupled to the underlying conductive feature (e.g., gate structure and/or silicide region **110** in the illustrated embodiment). The contacts **114** are electrically coupled to the gate structure and may be referred to as gate contacts, and the contacts **112** are electrically coupled to the silicide regions **110** and may be referred to as source/drain contacts. The barrier layer may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material **118** may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the second ILD **106**. The structure in the alignment mark region **10**A is illustrated in FIG. **28**D. Additional processes may be further performed, including back-end-of-line (BEOL) processes, scribing the substrate **10** to separate IC chips, more, which are not particularly illustrated herein.

[0079] FIGS. **29**A-C illustrate cross-sectional views of a device according to some alternative embodiments. FIG. **29**A illustrates reference cross-section A-A′ illustrated in FIG. **1**. FIG. **29**B illustrates reference cross-section B-B′ illustrated in FIG. **1**. FIG. **29**C illustrates reference cross-section C-C′ illustrated in FIG. **1**. In FIGS. **29**A-C, like reference numerals indicate like elements formed by like processes as the structure of FIGS. **28**A-C. However, in FIGS. **29**A-C, channel regions in the n-type region **10**N and the p-type region **10**P comprise a same material. For example, the second nanostructures **54**, which comprise silicon, provide channel regions for p-type nano-FETs in the p-type region **10**P and for n-type nano-FETs in the n-type region **10**N. The structure of FIGS. **29**A-C may be formed, for example, by removing the first nanostructures **52** from both the p-type region **10**P and the n-type region **10**N simultaneously; depositing the gate dielectric layers **100** and the gate electrodes **102**P (e.g., gate electrode suitable for a p-type nano-FET) around the second nanostructures **54** in the p-type region **10**P; and depositing the gate dielectric layers **100** and the gate electrodes **102**N (e.g., a gate electrode suitable for a n-type nano-FET) around the second nanostructures **54** in the n-type region **10**N. In such embodiments, materials of the epitaxial source/drain regions **92** may be different in the n-type region **10**N compared to the p-type region **10**P as explained above.

[0080] Embodiments may achieve advantages. For example, utilizing techniques described above such as hard masks for hot implantation processes, may help control the lateral boundaries of p-well **24** and n-well **38**, leading to a reduction of possible overlapping of the neighboring p-well **24** and n-well **38**, thereby improving the junction between the p-well **24** and n-well **38** and improving the leakage from the p-well **24** or n-well **38**.

[0081] In an embodiment, a semiconductor device includes a semiconductor substrate, the semiconductor substrate including a first fin and a second fin; a first well in the semiconductor substrate, wherein the first fin is in the first well, the first well being doped with a first dopant having a first conductivity type; a second well in the semiconductor substrate, wherein the second fin is in the second well, the second well being doped with a second dopant having a second conductivity type different from the first conductivity type; an alignment mark in the semiconductor substrate, wherein the alignment mark includes a lower portion formed of a first material and an upper portion formed of a second material different from the first material; and an isolation layer over the first fin, the second fin, the semiconductor substrate, and the alignment mark. In an embodiment, the first material is selected from the group consisting of silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof, and the second material is, different than the first material, selected from the group consisting of silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof. In an embodiment, the semiconductor device further includes a stack of semiconductor channel regions extending over the first fin. In an embodiment, the alignment mark has a top surface below a major upper surface of the semiconductor substrate. In an embodiment, the isolation layer extends below the major upper surface of the semiconductor substrate. In an embodiment, the first material is doped with the first dopant and the second dopant. In an embodiment, the second material is doped with the second dopant. In an embodiment, the first material is doped with the first dopant, and the second material is non-doped. In an embodiment, the first material and the second material are doped with only the second dopant.

[0082] In another embodiment, a method of forming a method is provided. The method includes forming a trench in a semiconductor substrate; forming a first patterned mask over the semiconductor substrate, wherein the first patterned mask includes a first material filling the trench; implanting a first dopant through the first patterned mask for forming a first well in the semiconductor substrate, the first dopant having a first conductivity type; removing the first patterned mask over a major upper surface of the semiconductor substrate and partially removing the first material in the trench; forming a second patterned mask over the semiconductor substrate, wherein the second patterned mask includes a second material filling the trench and covering the first material; implanting a second dopant through the second patterned mask for forming a second well in the semiconductor substrate, the second dopant having a second conductivity type being different from the first conductivity type; after the second well is formed, removing the second patterned mask over the major upper surface of the semiconductor substrate, wherein at least a portion of the second material remains in the trench after removing the second patterned mask; etching the semiconductor substrate to from a first fin and a second fin, wherein the first fin is in the first well, and the second fin is in the second well; and forming an isolation layer over the semiconductor substrate, wherein the isolation layer is along a sidewall of the first fin and a sidewall of the second fin and covering the second material in the trench. In an embodiment, the etching the semiconductor substrate to form the first fin and the second fin includes etching at least a portion of the second material in the trench. In an embodiment, the isolation layer fills the trench and is in contact with the second material. In an embodiment, the implanting the first dopant through the first patterned mask includes implanting the first dopant into the first material in the trench. In an embodiment, the implanting the second dopant through the second patterned mask includes implanting the second dopant into the first material and the second material in the trench. In an embodiment, the first material is selected from the group consisting of silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof. In an embodiment, the second material is different than the first material, selected from the group consisting of silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof.

[0083] In yet another embodiment, a method of forming a method is provided, and the method includes forming a trench in a substrate; forming a first hard mask over a major upper surface of the substrate and in the trench, wherein the first hard mask includes a first opening exposing a first region of the substrate; performing a first implantation process through the first opening of the first hard mask, to form a first well in the first region; etching the first hard mask while leaving a portion of the first hard mask in the trench; forming a second hard mask over the major upper surface of the substrate and over the portion of the first hard mask in the trench, wherein the second hard mask includes a second opening exposing a second region of the substrate; performing a second implantation process through the second opening of the second hard mask, to form a second well in the substrate, wherein the second well abuts the first well and has a conductivity opposite to the first well; etching the second hard mask to at least remove the second hard mask over the major upper surface of the substrate; depositing a stack of alternating first epitaxial layers and second epitaxial layers over the substrate and the second hard mask in the trench; and etching the stack and the substrate to form a first fin in the first well, a second fin in the second well, a first nanostructure stack over the first fin, and a second nanostructure stack over the second fin. In an embodiment, the second hard mask in the trench is etched while etching the stack and the substrate. In an embodiment, the method further includes forming an isolation layer adjacent the first fin, the second fin, and filling the trench. In an embodiment, the first hard mask and the second hard mask are formed of different materials.

[0084] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

## Claims

**1**. A semiconductor device comprising: a semiconductor substrate, the semiconductor substrate comprising a first fin and a second fin; a first well in the semiconductor substrate, wherein the first fin is in the first well, the first well being doped with a first dopant having a first conductivity type; a second well in the semiconductor substrate, wherein the second fin is in the second well, the second well being doped with a second dopant having a second conductivity type different from the first conductivity type; an alignment mark in the semiconductor substrate, wherein the alignment mark comprises a lower portion formed of a first material and an upper portion formed of a second material different from the first material; and an isolation layer over the first fin, the second fin, the semiconductor substrate, and the alignment mark.
**2**. The semiconductor device of claim 1, wherein the first material is selected from the group consisting of silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof, and the second material is, different than the first material, selected from the group consisting of silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof.
**3**. The semiconductor device of claim 1, further comprising a stack of semiconductor channel regions extending over the first fin.
**4**. The semiconductor device of claim 1, wherein the alignment mark has a top surface below a major upper surface of the semiconductor substrate.
**5**. The semiconductor device of claim 4, wherein the isolation layer extends below the major upper surface of the semiconductor substrate.
**6**. The semiconductor device of claim 1, wherein the first material is doped with the first dopant and the second dopant.
**7**. The semiconductor device of claim 6, wherein the second material is doped with the second dopant.
**8**. The semiconductor device of claim 1, wherein the first material is doped with the first dopant, and the second material is non-doped.
**9**. The semiconductor device of claim 1, wherein the first material and the second material are doped with only the second dopant.
**10**. A method of forming a semiconductor device, the method comprising: forming a trench in a semiconductor substrate; forming a first patterned mask over the semiconductor substrate, wherein the first patterned mask comprises a first material filling the trench; implanting a first dopant through the first patterned mask for forming a first well in the semiconductor substrate, the first dopant having a first conductivity type; removing the first patterned mask over a major upper surface of the semiconductor substrate and partially removing the first material in the trench; forming a second patterned mask over the semiconductor substrate, wherein the second patterned mask comprises a second material filling the trench and covering the first material; implanting a second dopant through the second patterned mask for forming a second well in the semiconductor substrate, the second dopant having a second conductivity type being different from the first conductivity type; after the second well is formed, removing the second patterned mask over the major upper surface of the semiconductor substrate, wherein at least a portion of the second material remains in the trench after removing the second patterned mask; etching the

semiconductor substrate to from a first fin and a second fin, wherein the first fin is in the first well, and the second fin is in the second well; and forming an isolation layer over the semiconductor substrate, wherein the isolation layer is along a sidewall of the first fin and a sidewall of the second fin and covering the second material in the trench.

**11**. The method of claim 10, wherein the etching the semiconductor substrate to form the first fin and the second fin comprises etching at least a portion of the second material in the trench.

**12**. The method of claim 11, wherein the isolation layer fills the trench and is in contact with the second material.

**13**. The method of claim 10, wherein the implanting the first dopant through the first patterned mask comprises implanting the first dopant into the first material in the trench.

**14**. The method of claim 13, wherein the implanting the second dopant through the second patterned mask comprises implanting the second dopant into the first material and the second material in the trench.

**15**. The method of claim 10, wherein the first material is selected from the group consisting of silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof.

**16**. The method of claim 10, wherein the second material is, different than the first material, selected from the group consisting of silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, and combinations thereof.

**17**. A method of forming a semiconductor device, the method comprising: forming a trench in a substrate; forming a first hard mask over a major upper surface of the substrate and in the trench, wherein the first hard mask comprises a first opening exposing a first region of the substrate; performing a first implantation process through the first opening of the first hard mask, to form a first well in the first region; etching the first hard mask while leaving a portion of the first hard mask in the trench; forming a second hard mask over the major upper surface of the substrate and over the portion of the first hard mask in the trench, wherein the second hard mask comprises a second opening exposing a second region of the substrate; performing a second implantation process through the second opening of the second hard mask, to form a second well in the substrate, wherein the second well abuts the first well and has a conductivity opposite to the first well; etching the second hard mask to at least remove the second hard mask over the major upper surface of the substrate; depositing a stack of alternating first epitaxial layers and second epitaxial layers over the substrate and the second hard mask in the trench; and etching the stack and the substrate to form a first fin in the first well, a second fin in the second well, a first nanostructure stack over the first fin, and a second nanostructure stack over the second fin.

**18**. The method of claim 17, wherein the second hard mask in the trench is etched while etching the stack and the substrate.

**19**. The method of claim 18, further comprising forming an isolation layer adjacent the first fin, the second fin, and filling the trench.

**20**. The method of claim 17, wherein the first hard mask and the second hard mask are formed of different materials.